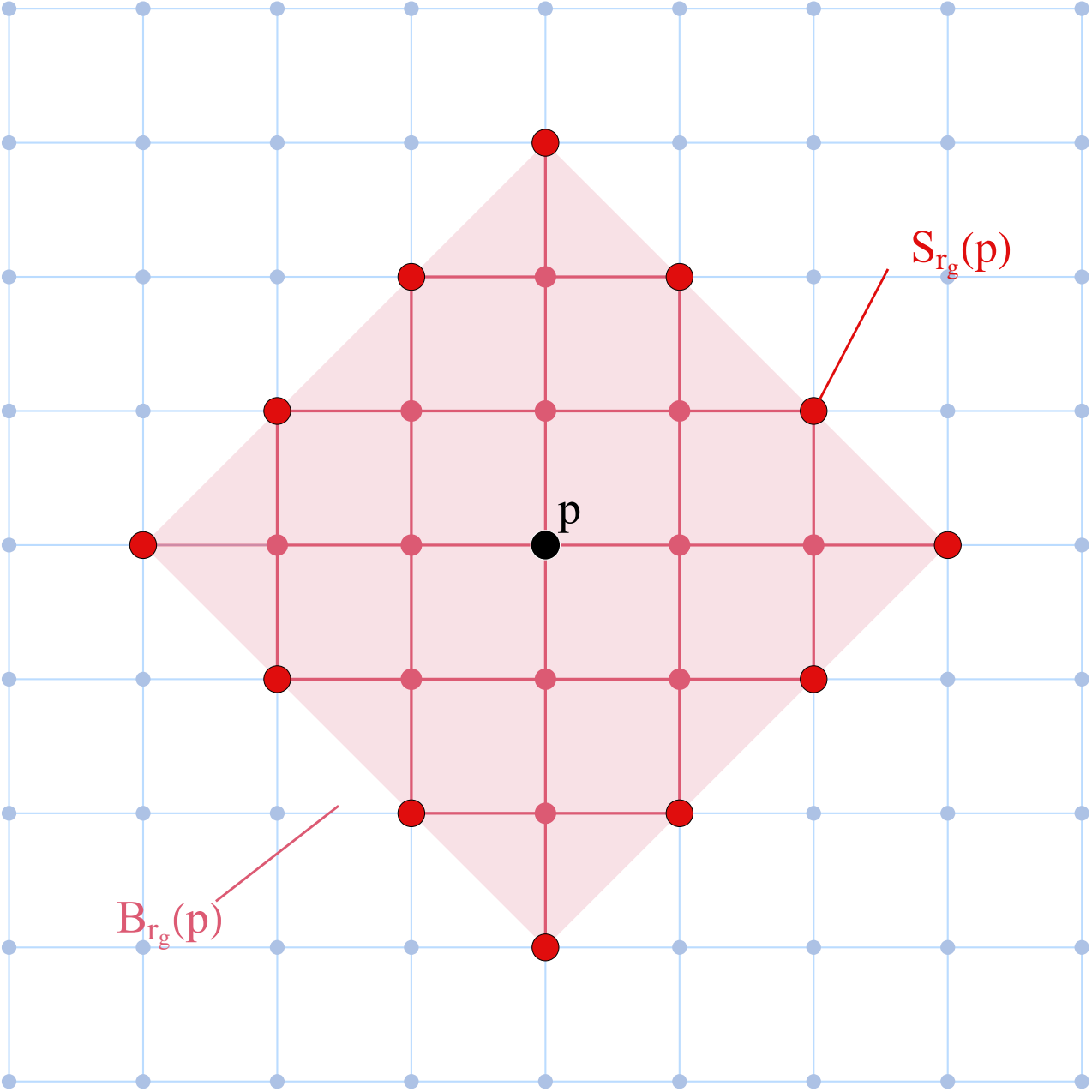
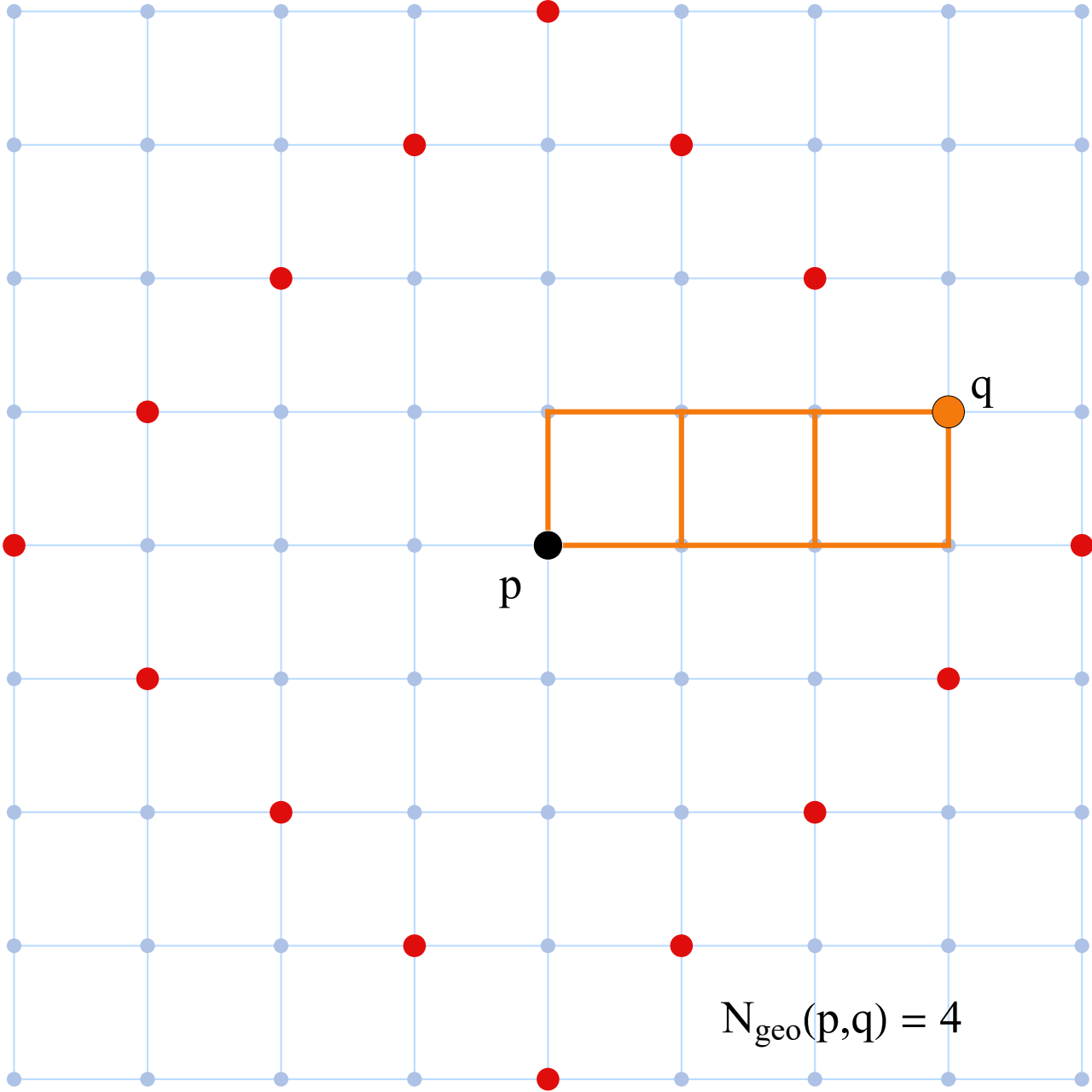


(a) Graph ball and shell



- $B_{r_g}(p) = \{ v \in V : d_G(p,v) \leq r_g \}$
- $S_{r_g}(p) = \{ v \in V : d_G(p,v) = r_g \}$

(b) Shortest-path multiplicities



collect $\log N_{geo}(p,q)$ for $q \in S_{r_g}(p)$